

Portugal's Economic and Human Development: A Century of Transformation

Task 3 — AID Project 2026

Group 29

1. Why are three separate fact tables used?

The three fact tables are kept separate because they represent different subject areas and, in the case of FACT_SECTOR, a different grain. FACT_ECONOMY and FACT_SOCIETY both contain one row per country and year, but they represent different groups of measures. FACT_ECONOMY contains economic measures such as GDP per capita, GDP, GDP growth and trade, while FACT_SOCIETY contains population and social measures such as GNI per capita, life expectancy and urbanization. FACT_SECTOR has a different grain: one row per country, year and sector. It contains sector-specific measures such as value added and employment. Keeping these facts separate therefore avoids mixing different grains in the same fact table and keeps each table focused on its subject area.

2. Why does DIM_SECTOR only apply to FACT_SECTOR?

DIM_SECTOR only applies to FACT_SECTOR because sector is part of the grain of that fact table. FACT_SECTOR contains one observation for each combination of country, year and sector, so each row must be associated with a sector from DIM_SECTOR. FACT_ECONOMY and FACT_SOCIETY, in contrast, have a grain of one row per country and year. Their measures are not separated by economic sector, so adding DIM_SECTOR to these fact tables would not provide a meaningful relationship and would introduce an unnecessary dimension.

3. How are the historical events integrated into the data warehouse?

In order to integrate the historical events dataset into the schema, we opted for the bridge table approach, linking each event to the DIM_COUNTRY and DIM_TIME tables. This approach preserves the grain of the fact tables (one row per country/year or one row per country/year/sector). It also allows multiple historical events to occur for the same country and year without duplicating fact rows. The bridge table contains one row per event, using a surrogate key together with foreign keys to DIM_COUNTRY and DIM_TIME.